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Poincaré Sections

From a flow to a return map

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MA4535 — Nonlinear Dynamics & Chaos | Lecture 7 | September 18, 2026

Outline

- **Flow**: advances states in time.
- **Trajectory** and **Orbit**: motion in time and its path.
- **Equilibrium Point**: a trajectory fixed at one state.
- **Poincaré Section**: a surface where trajectories cross transversely.
- **Return Map**: sends each crossing point to the next.
- **Periodic Orbit**: the closed path of a periodic trajectory.
- Rössler flow: plane and hilltop sections.

Flows

For an autonomous ODE $\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y})$ with state $\boldsymbol{y} \in \mathbb{R}^n$.

- The family φ_t is the **Flow**. It maps $\boldsymbol{y}_0$ to the state $\varphi_t(\boldsymbol{y}_0)$ at time t

$$\varphi_0(\boldsymbol{y}_0) = \boldsymbol{y}_0, \qquad \frac{d}{dt}\varphi_t(\boldsymbol{y}_0) = \boldsymbol{f}\big(\varphi_t(\boldsymbol{y}_0)\big).$$

- The solution $t \mapsto \varphi_t(\boldsymbol{y}_0)$ is a **Trajectory**. The path in state space $\mathbb{R}^n$ is the **Orbit**.
- An **Equilibrium Point** $\boldsymbol{q}_*$ satisfies $\boldsymbol{f}(\boldsymbol{q}_*) = \boldsymbol{0}$. The orbit is $\{\boldsymbol{q}_*\} \subset \mathbb{R}^n$ with constant trajectory $\boldsymbol{y}(t) = \boldsymbol{q}_*$.

Poincaré and Qualitative Dynamics

- In his 1881–82 memoirs, Poincaré studied orbit geometry without explicit trajectory formulas [10, 11]. He wrote, "One must naturally begin the study of any function with the qualitative part" (original in French) [10].
- Poincaré won the 1889 Prize Competition in Honour of King Oscar II with a qualitative study of the n-body problem. His memoir contained an error and did not solve the problem [14].
- In 1890, Poincaré tracked the three-body flow through successive crossings of a transverse surface. This is the origin of Poincaré sections [12].

Poincaré Sections

- A **Poincaré Section** Σ is a smooth surface used to record where trajectories cross

$$\Sigma = \{\boldsymbol{y} : h(\boldsymbol{y}) = 0\}, \qquad \nabla h \neq \boldsymbol{0} \quad \text{on } \Sigma.$$

  In three dimensions, Σ is a two-dimensional surface with normal ∇h.
  - Write $\Sigma^{\pm} = \{\boldsymbol{q} \in \Sigma : \pm\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) > 0\}$.
  - Trajectories $\boldsymbol{y}(t)$ satisfy $\partial_t h(\boldsymbol{y}(t)) = \nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) > 0$ as they cross $\boldsymbol{q} \in \Sigma^+$.
  - We focus on the positive crossings Σ^+ where trajectories go from $h < 0$ to $h > 0$.
  - If $\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) = 0$ the flow is tangent.
  - A **Transverse Crossing** has $\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) \neq 0$. It is tangent if $\nabla h(\boldsymbol{q}) \cdot \boldsymbol{f}(\boldsymbol{q}) = 0$.
- Poincaré studied sequences of points where a trajectory $\boldsymbol{y}(t)$ crossed Σ^+. **Crossing Points** of $\boldsymbol{y}(t)$ are all $\boldsymbol{q}_k = \boldsymbol{y}(t_k) \in \Sigma^+$ with $t_1 < t_2 < \cdots$.

First Return Map

Follow the trajectory $\boldsymbol{y}(t)$ from $\boldsymbol{y}(0) = \boldsymbol{q} \in \Sigma^+$ to its next crossing of Σ^+.

- **First Return Time** $\tau(\boldsymbol{q})$ gives the first positive time $\boldsymbol{y}(t)$ returns to Σ^+[1]

$$\tau(\boldsymbol{q}) = \inf\{t > 0 : \boldsymbol{y}(t) \in \Sigma^+\}.$$

  $D = \{\boldsymbol{q} \in \Sigma^+ : \tau(\boldsymbol{q}) < \infty\}$ contains the starting points that return.

- **First Return Map** P records the next crossing point

$$P(\boldsymbol{q}) = \boldsymbol{y}\big(\tau(\boldsymbol{q})\big), \qquad P : D \longrightarrow \Sigma^+.$$

  Successive crossing points satisfy $\boldsymbol{q}_{k+1} = P(\boldsymbol{q}_k)$ for $\boldsymbol{q}_k \in D$.

[1] Since $\inf \varnothing = +\infty$, $\tau(\boldsymbol{q}) = +\infty$ if the trajectory does not return.

Periodic Orbits

- A **Periodic Solution** satisfies, for some $T > 0$,

$$\boldsymbol{y}(t + T) = \boldsymbol{y}(t) \qquad \text{for every } t.$$

- For a nonconstant periodic solution, the smallest such T is the **Minimal Period**. Its **Periodic Orbit** is the closed path

$$\Gamma = \{\varphi_t(\boldsymbol{q}) : 0 \leq t < T\}.$$

- An equilibrium trajectory is constant. A periodic trajectory traces a closed orbit.

Periodic Points of Return Map

- A **Fixed Point** is a crossing point with $P(\boldsymbol{q}) = \boldsymbol{q}$. The flow trajectory returns to the same point at time $T = \tau(\boldsymbol{q})$.
- A **Periodic Point** of minimal map-period m satisfies

$$P^m(\boldsymbol{q}) = \boldsymbol{q}, \qquad P^j(\boldsymbol{q}) \neq \boldsymbol{q} \quad (1 \leq j < m).$$

- The flow trajectory returns to $\boldsymbol{q}$ after m crossings at time

$$T = \sum_{j=0}^{m-1} \tau\big(P^j(\boldsymbol{q})\big).$$

  - m counts crossings. T measures elapsed time.

Rössler Flow

- For $\boldsymbol{y} = (y_1, y_2, y_3)$, use

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}) = \begin{pmatrix} -y_2 - y_3 \\ y_1 + a y_2 \\ b + y_3(y_1 - c) \end{pmatrix}, \qquad a = b = 0.2, \quad c = 5.7.$$

- The first two coordinates rotate. The third lifts and folds each trajectory.
- Typical attractor trajectories do not repeat exactly [15].

Rössler Flow in Words

- When $y_3 \approx 0$, the top two equations approximate an outward spiral (y_1, y_2).
- When $y_1 > c$ and $y_3 > 0$, y_3 grows. The $-y_3$ term then pushes y_1 below c.
- Rössler described the return map this way [15]

  "The flow in state space allows for a 'folded' Poincaré map (horseshoe map)."

- Trajectories stretch and fold as they go over the hump. Trajectories never cross.

Plane Section $y_1 = 0$

- Set $h(\boldsymbol{y}) = y_1$ and record positive crossings of the plane $h = 0$

$$\Sigma = \{\boldsymbol{y} : y_1 = 0\}, \qquad \Sigma^+ = \{\boldsymbol{y} \in \Sigma : -y_2 - y_3 > 0\}.$$

- Each recorded vector $\boldsymbol{q}_k \in \Sigma^+$ satisfies $h(\boldsymbol{q}_k) = 0$ and $\nabla h(\boldsymbol{q}_k) \cdot \boldsymbol{f}(\boldsymbol{q}_k) > 0$.
- Return map P sends each recorded crossing point to the next

$$\boldsymbol{q}_k \longmapsto \boldsymbol{q}_{k+1} = P(\boldsymbol{q}_k).$$

Each $\boldsymbol{q}_k$ lies in $\mathbb{R}^3$, but $y_1 = 0$ leaves two free coordinates. In section coordinates, P maps a subset of $\mathbb{R}^2$ into $\mathbb{R}^2$.

Hilltop Section $y_3' = 0$

- The third Rössler equation gives $y_3' = f_3(\boldsymbol{y}) = b + y_3(y_1 - c)$. Set $h(\boldsymbol{y}) = f_3(\boldsymbol{y})$. Record negative crossings of $h = 0$ in the **Hilltop Section**

$$\Sigma = \{\boldsymbol{y} : h(\boldsymbol{y}) = 0,\ y_3(-y_2 - y_3) < 0\}.$$

- Each recorded $\boldsymbol{q}_k \in \Sigma$ is a negative crossing since, on $h = 0$, $\nabla h \cdot \boldsymbol{f} = y_3(-y_2 - y_3) < 0$.
  - Each $\boldsymbol{q}_k$ marks a nondegenerate maximum of y_3 with $y_3' = 0$ and $y_3'' < 0$.
  - The section is curved because $h = 0$ gives $y_3(y_1 - c) = -b$, which links y_1 to y_3.
  - Each $\boldsymbol{q}_k$ lies in $\mathbb{R}^3$, but $h = 0$ leaves two free coordinates. In section coordinates, the return map sends a subset of $\mathbb{R}^2$ into $\mathbb{R}^2$.

Hilltop Return Map $y_3' = 0$

- Record $\boldsymbol{q}_k \in \Sigma$ at each selected maximum

$$\boldsymbol{q}_{k+1} = P_{\text{top}}(\boldsymbol{q}_k), \qquad P_{\text{top}} : D_{\text{top}} \subseteq \Sigma \longrightarrow \Sigma.$$

- For $\boldsymbol{y} \in \Sigma$, $b = 0.2$ gives

$$y_3 \neq 0, \qquad y_1 = c - \frac{b}{y_3}.$$

  The curved surface has two free coordinates.

- Peak height does not determine the next peak. Both section coordinates matter.

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Historical Cast of Characters

From planetary observations to chaotic flows

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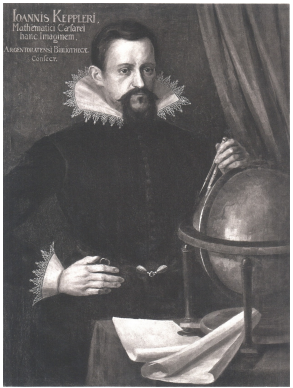
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Tycho Brahe (1546–1601)



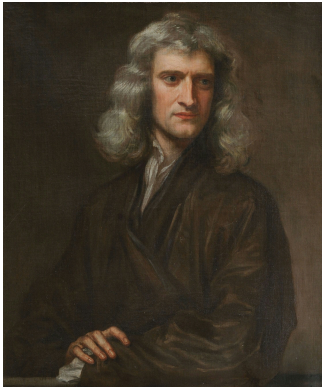
- Danish astronomer who made exceptionally precise observations before the telescope.
- Recorded the positions of Mars that Kepler later used to test planetary orbits.
- His measurements made small errors in circular models impossible to ignore.
- He also recorded the brilliant supernova of 1572.
- A duel cost him part of his nose; his prosthesis was likely brass.

Johannes Kepler (1571–1630)



- German astronomer and mathematician who worked with Brahe's observations.
- Found that Mars follows an ellipse, then formulated three laws of planetary motion.
- His laws described planetary motion accurately, setting a target for Newton's theory.
- An eight-arcminute mismatch in Mars's position helped convince him that circles would not do.
- He defended his mother when she was accused of witchcraft.

Isaac Newton (1642–1727)



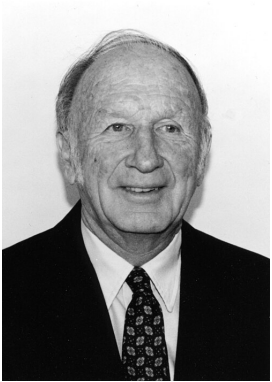
- English mathematician and physicist.
- In the 1687 *Principia*, gave laws of motion and universal gravitation.
- The same gravitational law explains falling bodies and Kepler's planetary orbits.
- He later ran the Royal Mint and pursued counterfeiters.
- He also kept extensive notes on alchemy.

Henri Poincaré (1854–1912)



- French mathematician who studied the geometry of trajectories without solving every equation explicitly.
- Investigated the three-body problem and the complex behavior of its orbits.
- Section crossings turn a continuous flow into a return map, the tool just used in this lecture.
- Correcting an error in his prize memoir revealed even more tangled three-body orbits.
- An insight once struck him as he stepped onto a carriage.

Edward Lorenz (1917–2008)



- American meteorologist at MIT who studied weather prediction with simplified equations.
- Showed that a deterministic model can amplify tiny differences in initial conditions.
- His 1963 work became a central example of chaos and limits to long-range prediction.
- A rerun with rounded input numbers produced a very different forecast.
- His famous 1972 talk asked whether a butterfly in Brazil could stir up a tornado in Texas.

Otto Rössler (born 1940)



- German scientist who proposed a compact three-variable chaotic flow in 1976.
- Its folded motion gives a clear setting for studying return maps.
- The Rössler flow supplied the plane and hilltop sections earlier in this lecture.
- His three-equation model needs only one nonlinear term.
- A taffy-pulling machine inspired its stretch-and-fold picture.

Summary

- A **Trajectory** is $\boldsymbol{y}(t)$. An **Orbit** is the trajectory's path in state space.
- A **Poincaré Section** is a surface. Trajectories cross sections at **Crossing Points**.
- The **First Return Map** sends each crossing point to the next.
- A **Periodic Orbit** is a closed path traced by a nonconstant periodic trajectory. A **Periodic Point** of the return map identifies a periodic orbit.
- Plane and hilltop sections are two of many possible sections for the Rössler flow.

History sources

- Brahe: NASA Earth Observatory on planetary observations, NASA on the 1572 supernova and his duel, and Kučera et al. on his prosthesis [7, 9, 8, 3].
- Kepler: NASA Earth Observatory and Owen Gingerich's biographical study, and the Stanford Encyclopedia of Philosophy [7, 2, 1].
- Newton: NASA Earth Observatory and The Royal Mint and the Newton Project's alchemical papers [7, 17, 16].
- Poincaré: the 1890 memoir and the King Oscar II prize history and Poincaré's *Science and Method* [12, 14, 13].
- Lorenz: MIT News on his work and the rounded-input rerun and his 1972 butterfly talk [5, 6].
- Rössler: *An Equation for Continuous Chaos* (1976) and Niklas Manz on the taffy machine [15, 4].

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- Rössler: Transmediale (2008), cropped by Chongkian. CC BY-SA 4.0.

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References I

[1] Daniel A. Di Liscia.
Johannes kepler.
Stanford Encyclopedia of Philosophy, 2025.

[2] Owen Gingerich.
Kepler.
Dictionary of Scientific Biography, reproduced by MacTutor History of Mathematics.

[3] J. Kucera et al.
Was he murdered or was he not? part ii: Multi-elemental analyses of hair and bone samples from tycho brahe and histopathology of his bones.
Archaeometry, 59(5):918–933, 2017.

[4] Niklas Manz.
Sabbatical trip to europe: Part 3 (otto rössler).
Wooster Physicists, 2024.

References II

[5] Massachusetts Institute of Technology.
Edward lorenz, father of chaos theory and butterfly effect, dies at 90.
MIT News, 2008.

[6] Massachusetts Institute of Technology.
Wind, war and weathermen.
MIT News, 2011.

[7] NASA Earth Observatory.
Planetary motion: The history of an idea that launched the scientific revolution.
NASA Science, 2009.

[8] NASA Goddard Space Flight Center.
Did you know? tycho brahe.
StarChild.

[9] NASA Science.
Stellar survivor from 1572 a.d. explosion supports supernova theory, 2004.

References III

[10] Henri Poincaré.
Mémoire sur les courbes définies par une èquation différentielle.
Journal de Mathématiques Pures et Appliquées, 7:375–422, 1881.

[11] Henri Poincaré.
Mémoire sur les courbes définies par une èquation différentielle.
Journal de Mathématiques Pures et Appliquées, 8:251–296, 1882.

[12] Henri Poincaré.
Sur le problème des trois corps et les èquations de la dynamique.
Acta Mathematica, 13:1–270, 1890.

[13] Henri Poincaré.
Science and Method.
Thomas Nelson and Sons, 1914.

References IV

[14] Mikael Rågstedt.
From order to chaos: The prize competition in honour of king oscar ii.
Institut Mittag-Leffler Library.

[15] Otto E. Rössler.
An equation for continuous chaos.
Physics Letters A, 57(5):397–398, 1976.

[16] The Newton Project.
Introducing newton's alchemical papers.

[17] The Royal Mint.
Sir isaac newton and the royal mint.